

Data & AI

8. Your Own Image Classifier

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Last Time

A classifier trained on somebody else's data

- 784 numbers in, 10 numbers out
- 50,890 parameters, trained in about a second
- Training accuracy 99.2%, test accuracy 95.0%

Today

- Assignment 1: an image classifier on images you collect
- Git, GitHub and GitHub Pages, because that is how you hand it in
- Google Colab, for those without a fast computer

What you will build

The Demo

espresso_cup · paper_coffee_cup · wine_glass

3 classes, 32x32 colour input, 393,731 parameters.

self test passed: the browser agrees with Python (largest gap 0.0000)

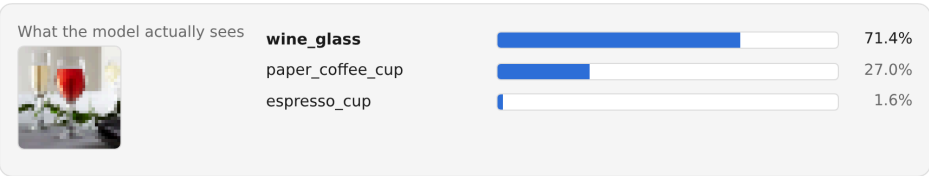
Webcam

Upload a picture

Draw it

Choose File0025.jpg

Pick a photo from your own phone or computer.



Built for MAS1004 Data & AI, Sogang University. The model is a small neural network trained on images collected by the student. It runs in JavaScript inside this page, so it works with no server at all. The forward pass is in `app.js` and is about forty lines long.

Three Ways to Feed It a Picture

- Your webcam
- A file you upload
- Something you draw with the mouse

It Runs Inside the Page

- No server, no Python, no installing anything
- The address keeps working after you close your laptop
- Nobody's photos are sent anywhere

Assignment 1

Train an image classifier on images you collect yourself

- 3 to 5 categories of your own choosing
- Due Friday 16 October, 22:00
- Everything at github.com/jdasam/mas1004-assignment1

The six problems

What You Are Graded On

	Problem	Points
1	Choose your categories and collect the images	20
2	Turn your folder into numbers	15
3	Train it and report the first result	20
4	Clean your data and train again	15
5	Test it on photographs you took yourself	20
6	Put the demo on the web	10

What You Are Given

- The image downloader
- The web page, with the webcam, upload and drawing already in it
- The export script and the checker
- A test for every function you have to write

What You Write

File	Functions
src/data.py	load_folder, split_train_test
src/train.py	build_model, train
src/evaluate.py	predict_logits, accuracy, confusion_matrix
src/evaluate.py	worst_examples

- Each one already has a description of exactly what goes in and what comes out

How You Know It Worked

```
pytest tests/test_data.py
```

- Red until you write it, green when it is right
- Paste the whole failure text to your agent, not your summary of it

The Plan Is Written Out for You

PLAN.md, sixteen steps, each with a way to check it

- Give it to your agent one step at a time, not all at once
- Read it first. Writing one of these is part of what you are learning

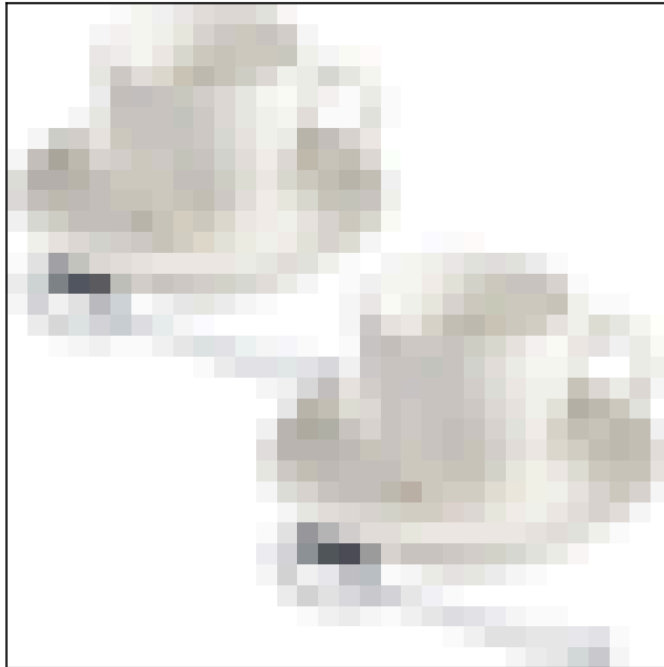
Choosing your categories

What the Model Actually Sees

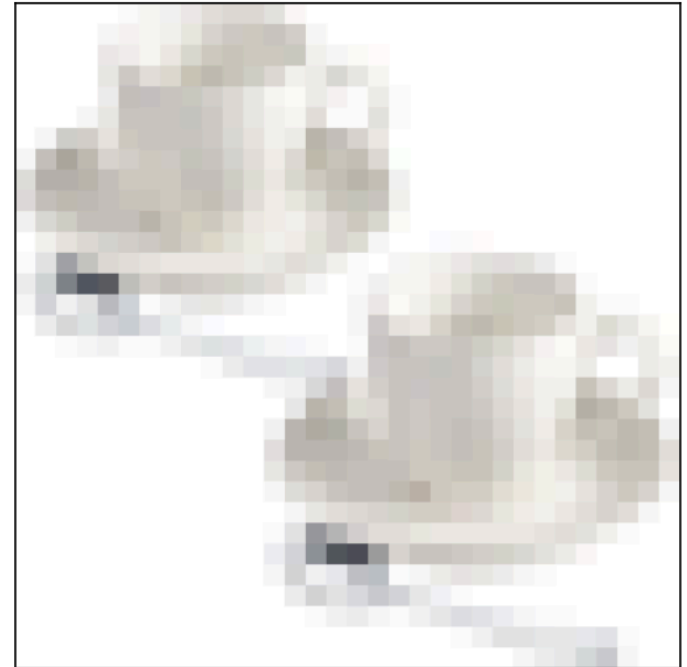
the photo
1000 x 1000 pixels



what you train on
32 x 32 pixels



the same 32 x 32,
blown up so you can see it



Categories That Work

- Told apart by colour and overall shape
- Three to five of them
- At least 50 images each, 150 is better

Pick Something You Care About

- The grade is in the explanation, not the accuracy
- You will look at these images one by one in Problem 4
- Choose something you would still find interesting at 60%

Downloading the Images

```
python src/collect.py --classes "espresso cup,wine glass,paper cup" --n 150
```

- About 70 seconds per category
- Search phrases, not category names: change the phrase if a category comes back empty

Your Test Set Is Not from the Internet

At least 5 photographs per category, taken by you

- Problem 5 measures the model on those, not on downloaded images
- It will do much worse on yours. Explaining that gap is the assignment

Git, GitHub, GitHub Pages

Why You Need These Today

- You hand in a repository address, not a zip file
- Your demo has to be on the web, and GitHub puts it there for free

Git

A program that records every version of your files

- Every save is kept, with a message saying what changed
- You can go back to any of them

GitHub

A website that holds git repositories

The screenshot shows the GitHub interface for the repository `jdasm/mas1004`. The top navigation bar includes links for Platform, Solutions, Resources, Open Source, Enterprise, and Pricing, along with search, sign in, and sign up buttons. The repository header shows the name `jdasm/mas1004` as public, with 10 stars and 1 fork. The main content area displays a list of files and folders: `data`, `demos`, `slide_decks`, `.gitignore`, `.nojekyll`, and `README.md`, each with a description and the time since the last update. The `README` section is expanded, showing the title `mas1004` and a description: "Data & AI at Sogang University. Materials for the 2026 course. Lecture slides are in `slide_decks/`. Interactive demos used in class (source in `demos/`, list of all demos at ...". The right sidebar contains sections for "About" (Data & AI at Sogang University), "Releases" (No releases published), and "Contributors" (2 contributors: jdasam and claude).

Platform Solutions Resources Open Source Enterprise Pricing

jdasm/mas1004 Public

Notifications Fork 1 Star 10

<> Code Issues Pull requests Actions Projects Security and quality Insights

2026 4 Branches 0 Tags

Go to file

<> Code

About

Data & AI at Sogang University

Readme

Activity

10 stars

1 watching

1 fork

Report repository

Releases

No releases published

Contributors 2

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claude Claude

README

mas1004

Data & AI at Sogang University

Materials for the 2026 course. Lecture slides are in `slide_decks/`.

Interactive demos used in class (source in `demos/`, list of all demos at ...)

Getting Your Own Copy

1. Make an account at github.com
2. Open github.com/jdasam/mas1004-assignment1
3. Press "Use this template", then "Create a new repository"
4. Keep it public

Three Commands

```
git clone https://github.com/<you>/<your repo>.git  
git add -A  
git commit -m "what I changed"  
git push
```

- Your coding agent can run all of these for you
- Push often. An unpushed commit is not handed in

GitHub Pages

Settings, then Pages, then Branch: main and Folder: /docs

- Your demo appears at `https://<you>.github.io/<your repo>/`
- It takes a minute or two the first time

Google Colab

A Computer in a Data Centre, for Free

- Nothing to install
- Notebook at `assignment1_colab.ipynb` in the repository

Everything You Save There Is Deleted

So keep the work in your Google Drive instead

```
from google.colab import drive  
drive.mount('/content/drive')
```

- The notebook then works inside My Drive/mas1004/assignment1

Turning On the GPU

Runtime, Change runtime type, T4 GPU, Save

- Do it before you run anything. The session restarts
- You do not need it for this assignment. Your model trains on a CPU in seconds

Now, in class

What to Get Done Before You Leave

1. A GitHub account
2. Your own copy of the repository, from the template
3. Your categories, written down
4. `collect.py` running

Before Thursday

- Take photographs of your own categories. At least 5 each
- Read `PLAN.md` and `README.md` all the way through
- Problems 1 and 2 finished

Coming Up

- Thursday: the training loop, opened up in PyTorch
- Then: images with more than one colour channel, and why 32 by 32 hurts

What would you want it to tell apart?